Solution Requirements

Date	04 july 2026
Team ID	xxxxxx
Project Name	Human Development Index(HDI) Predictor
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
1	Authentication	*Users securely log in using a username and password to access the HDI Prediction System.*	High
2	Authorization levels	*Data Analysts can upload datasets and generate predictions. Administrators can manage users, datasets, and system settings.*	High
3	External interfaces	*The system can import HDI-related datasets from sources such as WHO, World Bank, UNDP, or CSV/Excel files.*	High

4	Transactions processing	*The system validates uploaded data, preprocesses it, runs the machine learning model, and stores prediction results.*	High
5	Reporting	*The system generates prediction reports, charts, graphs, and downloadable CSV/PDF reports.*	Medium
6	Business rules, etc.	*The system predicts HDI only after successful data validation and preprocessing. Invalid or incomplete data is rejected.*	High
7	Compliance to laws or regulations	*The system protects user information and follows applicable data privacy and security guidelines.*	Medium
8	*Other*	*The system allows users to compare predicted HDI values with historical data and visualize trends using graphs and charts.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	*The system should quickly process uploaded datasets and generate predictions.*	*Prediction completed within **5 seconds** for a standard dataset.*
2	Scalability	*The system should support increasing amounts of data and multiple users.*	*Supports **1,000+ datasets** and **100 concurrent users**.*
3	Security & Data Privacy	*User data and prediction results must be protected using secure authentication and encryption.*	*Password encryption and secure HTTPS communication.*
4	Reliability & Availability	*The system should be available with minimal downtime and provide consistent prediction results.*	***99.9% uptime** per month.*
5	Usability & Accessibility	*The interface should be simple, user-friendly, and easy to navigate for analysts and policymakers.*	*Users can complete prediction tasks with minimal training.*
6	*Other*	*The system should allow easy updates to the machine learning model and future feature enhancements.*	Modular design with easy maintenance and updates.